

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458859.7947 +/- 0.0036 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.126 +/- 0.048
Transit depth (Rp/Rs)^2	1.59 +/- 1.22 [%]
Orbital Inclination (inc)	79.83 +/- 2.96
Airmass coefficient 1 (a1)	0.94 +/- 0.48
Airmass coefficient 2 (a2)	0.11 +/- 0.47
Scatter in the residuals of the lightcurve fit is	56.33 %
Best Comparison Star	#3 - [318, 59]
Optimal Aperture	0.75
Optimal Annulus	3.0
Transit Duration (day)	0.036 +/- 0.033

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.126 ± 0.048 vs 0.1368 (deviation 0.22σ)
Duration fit vs published	0.036 d vs 0.0758 d (deviation 1.2σ)
Mid-time O–C	-8.0 min
Depth SNR	2.6
Residual scatter	56.33 %

Light curve

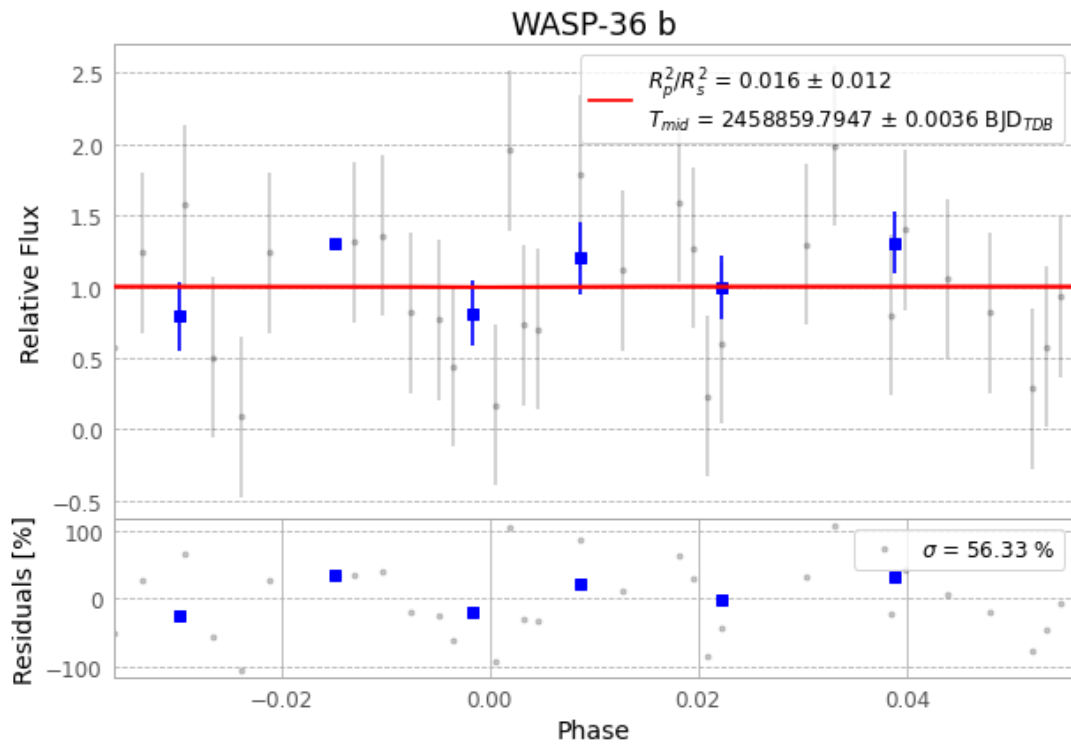


Figure 1 — FinalLightCurve_WASP-36 b_2020-01-10.png (SHA-256 254a1bef3b0dcc13...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [267, 409], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 a745260a720d4f1f...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

73 FITS frames (48 MB), WASP-36200111053012.FITS → WASP-36200111090612.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-36-200111.log (3d5d1e11c06e...), FinalLightCurve_WASP-36 b_2020-01-10.png (254a1bef3b0d...), FinalLightCurve_WASP-36 b_2020-01-10.csv (39da2f97a3aa...)

Compute resources

Wall time 180.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-03 03:52Z.